Day 1 Of Communication System Class

Communication is the process of establishing connection or link between two points for information exchange. Simply communication is the basic process of exchange information. The electronic equipments which are used for communication purpose are called communication equipments. Different communication equipments when assembled together form a communication system.

- <u>**COMMUNICATION MODEL (SYSTEM)**</u>

The purpose of a communications system is to exchange data between two entities.

- **Source:** entity that generates data; eg. a person who speaks into the phone, or a computer sending data to the modem.
- **Transmitter:** a device to transform/encode the signal generated by the source.

  - the transformed signal is actually sent over the transmission system.

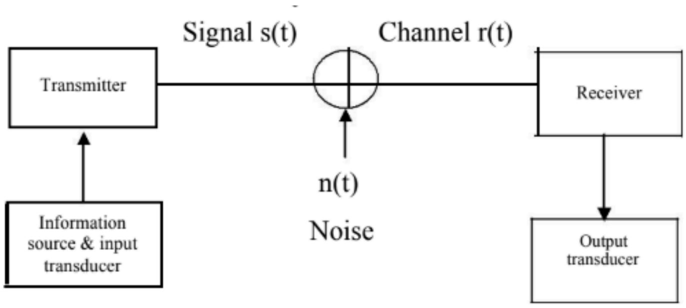
  eg. a modem transforms digital data to analog signal that can be handled by the telephone network.
- **Transmission System (Channel):** medium that allows the transfer of a signal from one point to another.

  eg. a telephone network for a computer/modem.
- **Receiver:** a device to decode the received signal for handling by destination device.

  eg. A modem converts the received analog data back to digital for the use by the computer.
- **Destination:** entity that finally uses the data.

  eg. Computer on other end of a receiving modem.

- **Basic Elements of a Communication System**
[Most Probable Question]



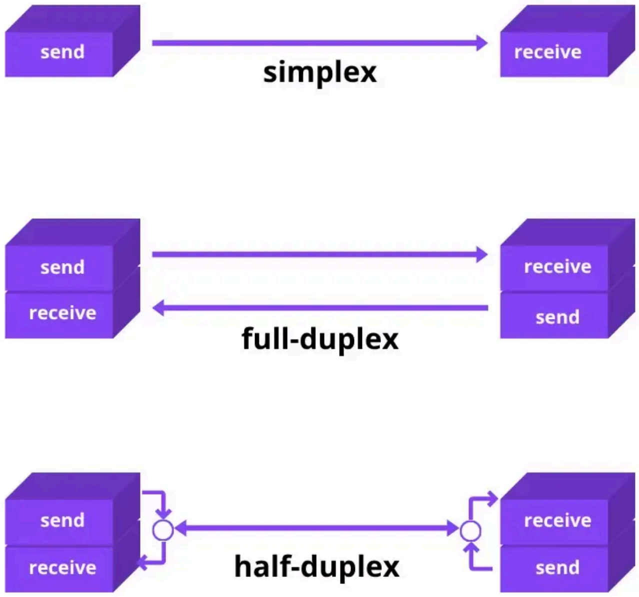
• **Information**: generated by the source may be in the form of voice, a picture or a plain text. An essential feature of any source that generates information is that its output is described in probabilistic terms; that is, the output is not deterministic. A transducer is usually required to convert the output of a source in an electrical signal that is suitable for transmission.

• **Transmitter:** a transmitter converts the electrical signal into a form that is suitable for transmission through the physical channel or transmission medium. In general, a transmitter performs the matching of the message signal to the channel by a process called modulation.
The choice of the type of modulation is based on several factors, such as:
- the amount of bandwidth allocated,
- the type of noise and interference that the signal encounters in transmission over the channel,
- and the electronic devices that are available for signal amplification prior to transmission.

• **Channel:** the communication channel is the physical medium that connects the transmitter to the receiver. The physical channel may be a pair of wires that carry the electrical signals, or an optical fibre that carries the information on a modulated light beam or free space at which the information-bearing signal are electromagnetic waves.

• **Receiver:** the function of a receiver is to recover the message signal contained in the received signal. The main operations performed by a receiver are demodulation, filtering and decoding.

Data flow *(simplex, half-duplex, and full-duplex)*



[Most Probable Question]

Q1. Compare digital and analog communication systems.

Digital Communication System	Analog Communication System
Advantages: • Inexpensive digital circuits • Privacy preserved (data encryption) • Can merge different data (voice, video and data) and transmit over a common digital transmission system	**Disadvantages:** • Expensive analog components: L and C • No privacy • Can not merge data from different sources • No error correction capability
Disadvantages: • Larger system Bandwidth • Synchronization problem is relatively difficult	**Advantages:** • Smaller bandwidth • Synchronization problem is relatively easier

Q2. Draw the generic block diagram of Analog communication system.

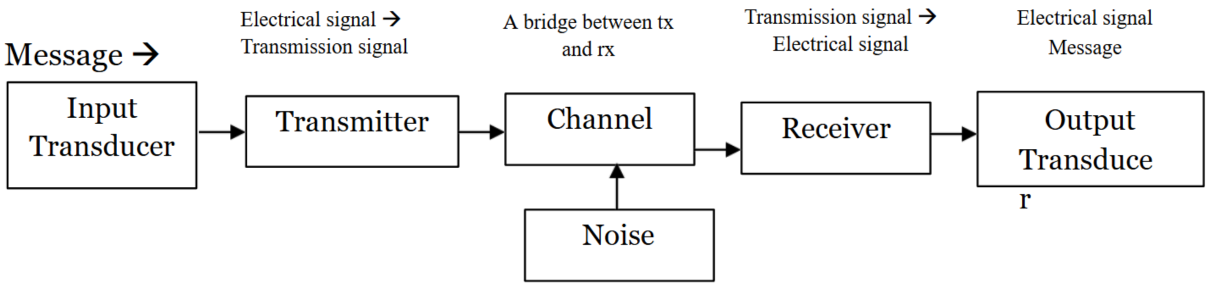


Figure: Block diagram of analog communication system

Q3. Draw the generic block diagram of Digital communication system.